Decentralized Multispectral UAV Swarms for Search and Rescue: Early-Fusion Perception, Delay-Tolerant Networking, and Energy-Aware Self-Organization

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Abstract. A large disaster area can be searched quickly only by many aircraft, and these must see people in poor light and keep working when one of them is lost. They must also share what they find over radio links that often fail. In this paper, we present a simulated multi-UAV search-and-rescue system that addresses these needs together. The system consists of a 4-channel RGB and thermal early-fusion YOLO11 person detector, a delay-tolerant ad-hoc mesh with store-carry-forward relaying of victim alerts, and a fully decentralized control layer in which every UAV plans from its own, possibly outdated, view of the swarm. The control layer combines energy-weighted centroidal Voronoi tessellation (EW-CVT) with dual-threshold hysteresis, sector claiming and lease-based failure suspicion, and it has no central planner. Each component can be switched off, and we measure the effect of doing so over 1,820 paired missions. Buffering alerts raises the share of victims known to the operator from 3.4% to 98% in a 5 km arena. Forwarding shortens the alert latency and helps when the finder is lost, and failover recovers 11 points of coverage after a UAV is lost. After a UAV is lost, the survivors need about 230 s on average to detect the loss and take over its sectors, because no UAV can know of the loss before the missing one is overdue at base. All results come from a kinematic simulation, and we discuss what this means for the conclusions.

Index Terms. UAV swarm, search and rescue, multispectral fusion, delay-tolerant networking, decentralized coverage control, centroidal Voronoi tessellation

I. Introduction

The first hours after an earthquake or a flood largely determine who is found alive. Speth et al. [6] note that the survival rate drops sharply with time, so a response that comes late helps few people. Small unmanned aerial vehicles (UAVs) can cover a wide area much faster than ground teams, and a swarm can cover it faster still. However, several practical problems stand in the way of using a swarm in a real disaster. Visible-light cameras perform poorly in shadow or smoke, where a thermal camera can still show a person. Over areas several kilometres wide the swarm is mostly out of radio contact, so a design that assumes a connected network would deliver almost nothing. A single ground planner would be a single point of failure and would need links that disaster areas rarely offer. If each UAV acts on its own, however, work may be duplicated or abandoned. This happens easily when the batteries differ and a UAV can be lost without anyone knowing.

We study perception, communication and control together, because each depends on the others. What a UAV can decide depends on what it has actually heard, and what it hears depends on the radio link and on where the other UAVs are. The contributions of this paper are as follows:

1) We build a 4-channel RGB and thermal early-fusion detector (YOLO11) and evaluate it under day, night and degraded-thermal conditions.
2) We design a delay-tolerant mesh in which victim alerts are buffered for each recipient and carried onward by any UAV that receives them (store-carry-forward), and we evaluate it with a physical radio model.
3) We design a decentralized control layer without a central planner. It applies an energy-weighted partition (EW-CVT) to the stale views of each UAV, and adds dual-threshold hysteresis, claim-and-yield sector assignment, and lease-based failure suspicion that tolerates long periods of silence.
4) We run a component-wise ablation over 1,820 paired simulated missions. We report the effects with confidence intervals, including the unfavourable ones, and we state what the simulation cannot show.

The rest of the paper is organized as follows. Section II reviews related work, and Section III gives an overview of the system. Section IV describes the methods and Section V the experimental setup. Section VI presents the results. Section VII discusses the limitations of the study, and Section VIII lists ongoing and future work. Section IX concludes the paper.

II. Related Work

UAVs for search and rescue. Small UAVs have been used to support wilderness and disaster search since the field trials of Goodrich et al. [19]. Recent surveys cover sensing and task assignment as well as path planning [18], and Cabreira et al. [20] review coverage path planning for a single UAV. For several robots, exploration and task allocation have been studied by Burgard et al. [21] and by Gerkey and Matarić [22], and auction-based methods such as CBBA [23] decide which robot searches which part of the area. Grøntved et al. [38] adapt CBBA to search

and rescue and assume a fully connected network, and Giacomossi et al. [39] re-auction the sectors of a failed UAV, but the failure is reported by the UAV itself and the simulation has no radio losses. These methods usually assume reliable communication, or they reach agreement by consensus [24]. Our setting is closer to flying ad hoc networks, in which the links are sparse and intermittent [25], [26].

Multispectral person detection. Combining visible and thermal images is known to improve pedestrian detection in poor light [1], [2], [3], [28]. Early fusion, in which the two modalities are stacked as input channels, needs little extra computation and is therefore suitable for onboard hardware [6]. We use it on top of the YOLO family [4], [5], [29]. Speth et al. [6] applied RGB and thermal early fusion to person detection from a drone and released the multispectral dataset that we use, and our detector follows their early-fusion design.

Coverage control. Dividing an area among agents with Voronoi cells and moving each agent to the centroid of its cell (Lloyd iteration) is a standard coverage method [7], [8], [9]. Versions that need only limited-range interaction between agents have also been proposed [10], as have adaptive and fully decentralized control laws [30]. In contrast to these works, our agents act on outdated views received over a lossy link and resolve conflicts by claiming sectors. Weighting the distance by a quantity that belongs to each agent gives a weighted Voronoi diagram. A power diagram [11] subtracts the weight from the squared distance, and we divide by it instead. Energy-aware versions of Lloyd's method use such weights to give smaller regions to robots with less energy [36], [37]. Our weight represents the battery level in the same way, so we do not claim the weighting itself as new. The weight rule of Munir et al. [37] reduces to our case $\alpha = 1$ when all UAVs drain at similar rates, so we do not run it separately. We add hysteresis to the updates to avoid chattering when the inputs are noisy.

Failure detection. Deciding that a peer has failed when the channel is unreliable is the problem addressed by unreliable failure detectors [33]. Accrual detectors output a suspicion level that is computed from the arrival times of heartbeats [34]. This is not suitable here, because silence is the normal state of a disconnected mesh. Instead, our sortie lease follows the lease idea of Gray and Cheriton [35]. A claim is valid until a stated time, and after that time it can be released.

Delay-tolerant networking. When end-to-end paths rarely exist, nodes can store messages and forward them when they meet other nodes [12]. Epidemic and spray-and-wait routing [13], [14] trade the delivery rate against the number of copies and the buffer use, probabilistic variants use the contact history [31], and gossip-style dissemination follows a similar idea [32]. GeoSaW [40] forwards alerts of search-and-rescue UAVs to carriers whose planned waypoints pass the destination. We use epidemic forwarding with a buffer cap and do not use planned routes, and Section VI-E compares it with spray-and-wait, PRoPHET and a GeoSaW-style forwarder in the same simulator. Simulators of UAV networks couple flight and network models [16], and our radio model is a simple log-distance channel [17]. Air-to-ground links can be better than this model when a line of sight is likely [27], so our model is probably conservative. Our per-node broker follows the topic-subscription idea of distributed MQTT brokers with data-link look-up (DMLT) [15]. In the mission simulator all alerts share one topic, so only the per-recipient buffering of the broker is exercised and topic filtering is not.

III. System Overview

Fig. 1 shows the system. A team of N UAVs launches from a common base and searches an $L \times L$ arena divided into sectors. Every UAV runs the same three components. Perception turns a multispectral frame of the sector below into victim detections. Mesh communication sends alerts and periodic status messages over a lossy radio link, buffers what could not be sent, and carries received alerts onward. Control chooses which sectors to search, from the UAV's own battery and its current view of its peers, and builds a route. The ground control station (GCS) is a passive sink: it receives alerts and plans nothing.

The interfaces are narrow. Detections go to the mesh as alerts, status messages from the mesh update the view that the control layer has of its peers, and control hands a route to flight. The detection, networking and control logic are the actual code under test, whereas the aircraft and the radio are simulated.

IV. Method

A. Early-fusion perception

The visible (RGB) and thermal long-wave infrared frames are stacked into a four-plane tensor, and the first convolution of the detector takes four input channels. The shipped model is YOLO11s with an input size of 1280 px, and the frames are 1280×937. It is trained on the NII-CU Multispectral Aerial Person Detection dataset [6], with a single person class. We initialise the thermal kernel with the pretrained red-channel kernel, and we train without colour jitter because it has no meaning for a thermal plane. The box centre of a detection is mapped linearly into the bounds of the sector in which the frame was taken, which gives a world coordinate. This is sector-based georeferencing. It is not a camera model.

B. Radio channel and delay-tolerant mesh

Channel. The path loss between two UAVs at distance d is modelled by Eq. 1.

$$PL(d) = PL(d_0) + 10\, n \log_{10} \tfrac{d}{d_0} + X_\sigma + \Delta_{\text{obs}}, \quad (1)$$

where $\Delta_{\text{obs}} = 15\,\text{dB}$ when the link is obstructed (a random 10% chance beyond 50 m) and 0 otherwise, with

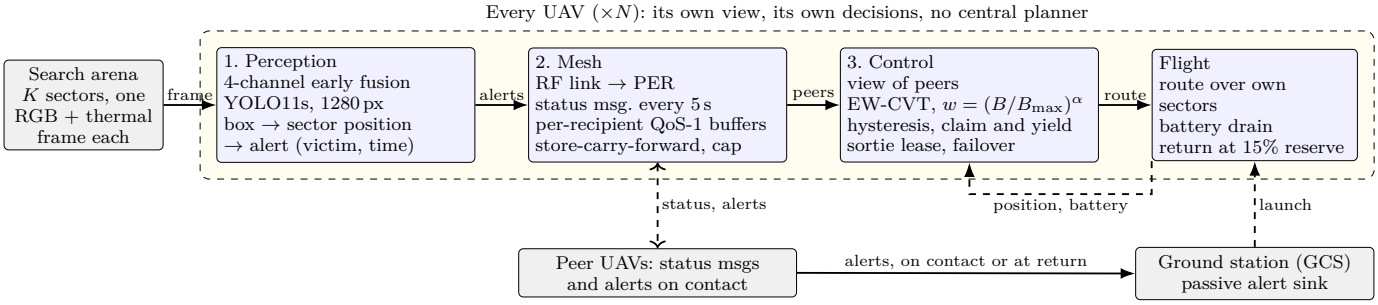


Fig. 1. System architecture of one UAV and the ground station.

$PL(d_0) = 20\log_{10} f_{\mathrm{MHz}} - 27.55$, $n = 2.5$, $d_0 = 1\,\mathrm{m}$, log-normal shadowing X_σ with $\sigma = 4\,\mathrm{dB}$, and a noise floor of $-90\,\mathrm{dBm}$. $\mathrm{SNR} = P_{tx} - PL - N_0$ maps to a packet error rate: $\mathrm{PER} = 0$ above 25 dB, 1 below 5 dB, and $1 - (1 + e^{-0.5(\mathrm{SNR}-15)})^{-1}$ between. Arenas of 2 km or more use 33 dBm at 2.4 GHz, giving $\mathrm{PER} \approx 0.5$ at about 520 m; smaller arenas use 30 dBm at 5 GHz (about 220 m). Against arenas of several kilometres, the swarm is mostly disconnected in flight.

Alerts. When a UAV detects a victim, it makes one direct attempt to reach every other UAV and the GCS. Each failed copy is stored in a per-recipient QoS-1 buffer. The buffer is flushed when the link to that recipient has a PER below 0.2, which in practice happens when the UAV returns to base.

Store-carry-forward. A UAV that receives an alert also keeps a copy for each recipient that is not known to hold it, and it hands the copy over when it later meets that recipient with a good link. Therefore, no UAV needs to be at the right place at the right moment, and an alert survives the loss of the UAV that found the victim. On contact, only the copies that the other side lacks are sent. When the buffer of a UAV is full, the oldest copy is dropped.

Status messages. Every 5 s, each UAV broadcasts its position, its committed EW-CVT generator and weight, its state, the time at which its route should bring it home, its sector claims, and the set of sectors that it knows to be scanned. These messages are single-hop and are not buffered, since stale state is not worth delivering late. A receiver merges the scanned sets as grow-only sets, and thus the knowledge about coverage spreads transitively when UAVs meet.

C. Decentralized energy-aware control

EW-CVT partitioning. Let B_i be the battery level of UAV i and $B_{\max}$ the capacity. The UAV has the weight $w_i = (B_i/B_{\max})^\alpha$. A sector with centre c_s is preferred by the UAV with the smallest weighted distance

$$d(s,i) = \frac{\|c_s - g_i\|^2}{\max(0.01,\, w_i)}, \tag{2}$$

where g_i is the committed generator of the UAV. For $\alpha = 0$ the partition is a plain Voronoi diagram, and a larger α shrinks the territory of low-battery UAVs. The launch partition is the converged CVT (15 Lloyd iterations) for the launch batteries.

Dual-threshold hysteresis. Every 5 s, each UAV computes the centroid C_i^* that its cell should have. It does so over the peers that it believes to be active, using their last advertised generators and weights. The UAV commits the new generator and weight only if

$$U_i = \big(\Delta B_i \geq \tau_\epsilon \;\wedge\; \|C_i^* - C_i\| \geq \tau_\delta\big) \;\vee\; \big(\mathcal{N}_i(t) \neq \mathcal{N}_i(t_k)\big), \tag{3}$$

with $\tau_\epsilon = 2\%$ of $B_{\max}$, $\tau_\delta = 5\,\mathrm{m}$, and $\mathcal{N}_i$ the view of the live set held by the UAV. Battery readings contain sensor noise, so without this gate every tick would commit a new partition and the borders would jitter.

Claims. A UAV claims a sector if it is the preferred owner by $d(s,i)$ (Eq. 2) using committed values and no other UAV holds a valid claim on the sector. It keeps its existing claims unless a better claim (a smaller d, with ties broken by id) reaches it. It may take over the claim of a peer only if it would arrive well before the advertised arrival time of that peer. A UAV that has no claims left takes the nearest unclaimed sector. Routes are greedy nearest-neighbour tours that end at the launch pad.

Failure suspicion by lease. In flight, silence tells a UAV nothing about a peer, because the peer may be out of range or it may have landed. Instead, each peer advertises the time at which its route should bring it home. A peer that is still missing slack seconds after this time (60 s by default) is suspected to be lost, and its claims are released. A UAV in flight cannot distinguish a lost peer from one that has landed quietly. Therefore, released work is taken only by UAVs at base, which have heard the landed peers. Landed UAVs listen for 10 s, split the work among themselves by $d(s,i)$, and relaunch if the battery allows. A suspicion is withdrawn when the peer is heard again. A UAV aborts and returns at 15% battery, and it is lost at 0%.

V. Experimental Setup

Missions. Six UAVs search a $5 \times 5\,\mathrm{km}$ arena that is divided into 25 sectors. Each sector shows one frame of the NII-CU multispectral aerial person detection dataset [6], and the labels of the frame serve as ground truth (36

Algorithm 1 Control tick of one UAV (every 5 s)

1: merge any status messages received since the last tick into the view (peers, claims, scanned set)
2: for each peer j overdue at base by more than slack: mark j suspected; release its claims
3: $\mathcal{N}_i \leftarrow$ self $\cup$ peers believed active (landed peers included only if self is at base)
4: read own battery B_i (noisy); $w_i^* \leftarrow (B_i/B_{\max})^\alpha$
5: $C_i^* \leftarrow$ EW-CVT centroid of my cell over $\mathcal{N}_i$ using peers' advertised (g_j, w_j)
6: if U_i (Eq. 3) then commit $g_i \leftarrow C_i^*$, $w_i \leftarrow w_i^*$
7: keep claims not beaten by a better claim; add free sectors I prefer by $d(s,i)$
8: if no claims left: claim nearest unclaimed sector; if at base: also share released work
9: if claims changed: rebuild nearest-neighbour route ending at the pad
10: advertise route ETA, claims and scanned set in the next status message

to 56 victims per world). The UAVs launch from a pad, fly at up to 50 m/s in transit and 32 m/s while searching, and use 0.07%/s of battery above 40 m/s and 0.04%/s otherwise. The launch batteries are uniform between 20% and 100%, and the battery readings have a noise of $\sigma = 1.5$ points. In the loss scenario, UAV 2 fails silently at $t = 80$ s and the other UAVs are not told.

Seeds and statistics. A seed fixes the frame in each sector and every launch battery, so each variant sees the same 20 worlds. We report paired differences with 95% confidence intervals, and we do not claim an effect when an interval contains zero. Missions run until completion or 1,500 s. Since each sector shows one static image, the detections inside missions are cached per frame.

Metrics. Coverage is the share of sectors that were scanned at least once. Precision and recall are computed by matching reported victims to the ground truth within 15 m (one-to-one), and recall is taken over all victims in the arena. Operator recall counts only the victims whose alert reached the GCS, and alert latency is the delay between detection and arrival at the GCS. Failover latency is the time from the injected failure to the first claim of one of its sectors by a survivor. Duplicate scans count the sectors that were scanned more than once. Sector flips count the claims that a UAV gave up to a better claim, route changes count the route rebuilds, and commits count the updates approved by the hysteresis.

Baselines. Each baseline replaces one policy in the same simulator, so the missions, radio model and seeds are identical. For forwarding, spray-and-wait [14] gives each alert $L = 3$ copies and halves them at each hand-over. PRoPHET [31] hands a copy to a peer with a higher delivery predictability for the base ($P_{\text{init}} = 0.75$, $\beta = 0.25$, $\gamma = 0.98$ per 30 s). The GeoSaW-style forwarder keeps one copy and gives it to a peer whose return time to base is more than 10 s shorter. This only approximates GeoSaW [40], which uses planned waypoints, and it is not the original protocol. Buffer only carries each alert home with its finder. For allocation, the auction baseline assigns each sector to the nearest UAV (all weights equal to 1) and keeps our lease-based failover. It is a nearest-bidder rule and not a full CBBA [23]. We also test it without failover and a static partition. Copies per alert counts the UAVs that hold a copy at the end.

TABLE I
Detection results on 150 held-out frames.

Condition	Input	AP50	AP75	Prec.	Rec.
Day	Fused	99.8	85.8	0.993	0.989
Day	Thermal only	89.0	50.5	0.944	0.731
Night	Fused	94.3	58.3	0.916	0.869
Night	Thermal only	89.0	50.5	0.944	0.731
Washout	Fused	61.8	25.1	0.991	0.386
Washout	Thermal only	0.0	0.0	0.000	0.000

Detector evaluation. We run the shipped model on 150 held-out validation frames (IoU 0.5, confidence 0.25). In the night condition, the RGB plane is darkened to 12% brightness and noise is added. In the thermal washout condition, the thermal contrast is scaled to 25%. In the RGB-only and thermal-only conditions, the other plane is set to zero. Note that this is modality-masked inference of one fused network and not a comparison of separately trained models.

Reproducibility. All numbers are produced by scripts in the project repository. A detector evaluation and an ablation harness produce the data, and the harness runs all 1,820 paired missions in a few minutes once the detections are cached. A separate report generator writes the tables and figures. Every component is a command-line switch of the mission simulator.

VI. Results

A. Perception

Table I shows that the fused model reaches an AP50 of 99.8 in daylight. Fusion is about 11 points better than thermal only, which suggests that the visible plane adds information. Removing the visible light costs only 5.5 points, because thermal carries most of the detection. Degraded thermal contrast is the hardest case, where the recall falls to 0.39 while the precision stays at 0.99. The RGB-only row is zero because the network was never trained without the thermal plane. It therefore says nothing about the value of RGB, and trained single-modality baselines are still needed (Section VIII).

B. Reference mission

Without a failure, the decentralized swarm covers 100% of the arena in 353±31 s (mean ± s.d. over 20 seeds). Precision, recall and F1 are all 0.980±0.015, where the whole-arena recall counts the victims in unsearched sectors. The mean localization error is 1.45±0.15 m (RMSE 1.86±0.32 m), the first victim is detected after 8.6±7.2 s, the UAVs use 13.7±1.4% of their battery on average, and the operator learns of 100% of the reports. Because each sector is one static frame, the localization error measures the mapping from box to sector and not geolocation.

TABLE II
Ablation results (paired mean ± 95% CI, 20 seeds).

Component removed: measure	Full	Ablated	Change
Failover (UAV lost): coverage	98.4%	87.4%	-11.0 ± 2.6
whole-arena recall	0.962	0.840	-0.122 ± 0.032
Hysteresis†: sector flips	2.5	13.5	$+11.0 \pm 3.5$
route changes	4.8	25.4	$+20.6 \pm 6.4$
mission time	362 s	392 s	$+30 \pm 25$ s
EW weighting (α=0): min. battery	18.8%	17.9%	-0.8 ± 0.7
DTN buffer (5 km): operator knows	98.0%	3.4%	-94.7 ± 1.2
swarm-wide consensus	100%	22.1%	-77.9 ± 1.3
Forwarding (5 km): alert latency	103 s	109 s	$+7 \pm 2$ s

† test configuration with global recomputation, $\sigma = 2\%$ (see text).

TABLE III
Forwarding baselines at 3 km (mean of 20 seeds).

Policy	Lat. (s)	Lat., loss (s)	Recall, loss	Copies
Epidemic (ours)	41.0	46.3	0.978	4.30
Spray-and-wait	48.6	56.4	0.967	2.37
PRoPHET	51.8	56.4	0.963	3.19
GeoSaW-style	47.9	56.0	0.973	1.63
Buffer only	59.2	66.4	0.958	1.00

C. What each component buys

Table II gives the paired effect of removing each component.

Failover matters when a UAV is lost. The coverage after the loss is 98.4% with failover and 87.4% without it (Fig. 3), since without failover the sectors of the lost UAV are never searched.

Hysteresis reduces churn but not flight time. In the full decentralized system, it has almost no measurable effect, because the claims are already sticky (0.2 versus 0.4 sector flips). It matters when the partitions are recomputed from fresh battery readings at every tick without this stickiness. We tested this in a configuration in which one planner with global knowledge recomputes all partitions at every tick. There, hysteresis cuts the number of partition commits about tenfold (296 to 31 per mission at $\sigma = 2\%$) and the number of sector flips from 13.5 to 2.5, while the change in mission time is within noise. We therefore see hysteresis as a protection against noisy inputs and not as a gain in speed.

EW weighting is a trade-off. When α is swept from 0 to 3, the minimum final battery rises from 15.6% to 20.3% and the low-battery aborts fall from 0.95 to 0.30 per mission. However, the mission becomes longer, from 377 s to 488 s, because the healthy UAVs carry more of the work. At $\alpha = 1$ the benefit is modest ($+0.8 \pm 0.7$ points of minimum battery).

D. Networking

At these ranges, the first-attempt delivery rate to the GCS is only 3 to 6%, which means that the swarm is mostly disconnected in flight (Fig. 4). Without a delay-tolerant buffer, the operator learns of 3.4% of the victims at 5 km and the swarm-wide consensus is 22%. With the buffer, these values are 98% and 100%. Buffering dominates every other effect that we measured.

Store-carry-forward lowers the alert latency (mean 41 versus 59 s at 3 km, 34 versus 41 s at 1.5 km, and 103 versus 109 s at 5 km). It also protects alerts when the finder is lost. With a UAV lost at 60 s, the operator learns of 99.3% of the reports at 3 km with forwarding and 97.3% without it (98.9% versus 97.6% at 1.5 km). Forwarding needs buffer space, however. When the buffer is capped at 5, 15, 40 and 120 copies per UAV, the operator learns of 22.5%, 41.5%, 83.4% and 99.3% of the reports (3 km, UAV lost). The benefit is real. It also has a price in memory.

E. Comparison with baselines

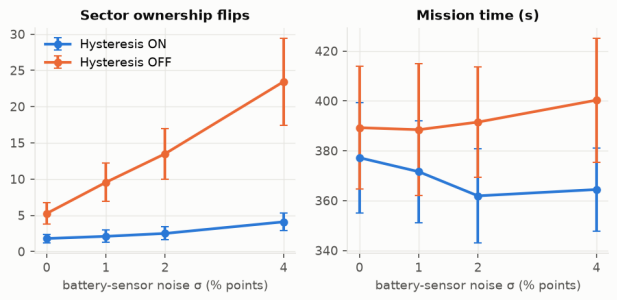
Table III compares the forwarding policies. Epidemic forwarding has the lowest alert latency in every scenario, but it also holds the most copies. At 3 km, its latency is 7.6±1.7 s below spray-and-wait, 10.7±3.2 s below PRoPHET and 6.8±2.4 s below the GeoSaW-style forwarder (paired differences with 95% confidence intervals), and it keeps 1.1 to 2.7 more copies per alert. When UAV 2 is lost, the operator recall is 0.978 with epidemic forwarding and 0.958 to 0.973 with the others, and the latency gap grows to about 10 s. Otherwise the operator recall is the same for all policies.

Against nearest-bidder allocation, the picture is mixed. Without a loss, energy weighting leaves a higher minimum battery than nearest-bidder allocation (+1.4±1.1 points) and fewer duplicate scans (−1.15±0.53 per mission), but it is not faster (+13±19 s, which includes zero). After a loss it is slower (554 versus 481 s, +73±58 s), and the coverage does not differ significantly (+1.4±3.2 points). The large gain after a loss comes from the failover rule. Without it, the coverage falls from 98.4% to 89.2% and the arena recall from 0.96 to 0.87, and a static partition ends at 87.6% (mean of 20 seeds). We therefore credit the failover with the coverage gain, and the weighting only with the energy reserve and the duplicate scans.

F. After a UAV loss

Table IV shows the behaviour of the swarm when UAV 2 is lost. The survivors cannot know that the UAV is gone until it is overdue at base, so failover takes minutes. It takes 231 s on average at 5 km, 143 s at 3 km and 102 s at 1.5 km, and the shorter times in smaller arenas follow from the UAVs meeting more often. The coverage stays at 98.4% or above, but the mission at 5 km grows from 353 s to 554 s. In a few seeds, some sectors stay unsearched because the survivors no longer have the battery for a second sortie when they learn of the loss (for instance, when the lost UAV carried the most charge).

Lease slack. The slack trades the speed of detection against false alarms. For slack values of 20, 60, 120 and 240 s, the mean failover latency is 192, 231, 293 and 411 s,



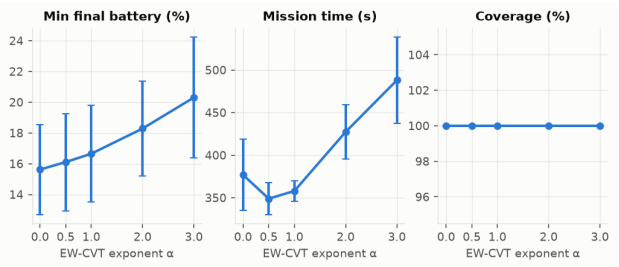


Fig. 2. Left: hysteresis versus sensor noise (global-recomputation test configuration). Right: effect of α.

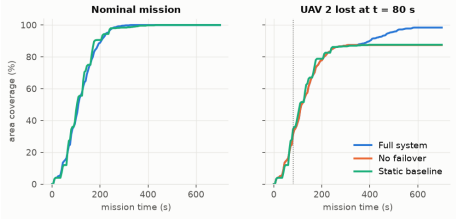


Fig. 3. Coverage over time without a failure (left) and with UAV 2 lost at 80 s (right).

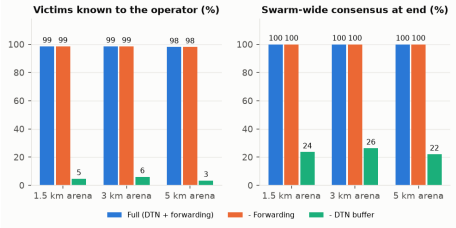


Fig. 4. Victims known to the operator (left) and swarm-wide consensus (right).

and the number of wrong suspicions falls from 11.2 out of 15.6 to 0.4 out of 4.7 per mission. Most suspicions concern UAVs that were fine, since a UAV that has landed looks like one that has crashed to any peer that is still in flight. These false suspicions cost little because released work is taken only at base, and the duplicate scans stay at about one per mission. The interval between status messages has little effect on the mission time (532, 554, 587 and 629 s at 2, 5, 15 and 60 s) until it becomes comparable to the duration of a contact.

VII. Discussion and Limitations

Our evidence supports a few conclusions clearly. A sparsely connected swarm needs buffering. Losing a UAV costs coverage unless the others take over its work, and the time that this takes is set by how long the survivors need to learn of the loss. Other results are weaker than the design might lead a reader to expect. Hysteresis has an effect only where the partitions are recomputed from noisy readings at every tick, and energy weighting buys reserve at the price of speed.

We observed several limitations in our study. First, the simulator is kinematic. It uses constant speeds and a linear battery drain, and it has no wind and no obstacles other than a probabilistic radio penalty. Second, each sector shows one static dataset frame. The mission-level precision and recall therefore mostly reflect the detector on these frames and the sectors that were visited, and the localization error measures a mapping from box to sector. Third, the perception ablation masks one plane of a single fused network, and the training and validation frames come from the same site, so the absolute numbers are probably optimistic. Fourth, the network results depend on the radio model. Fifth, the decentralized design makes several simplifications. The UAVs launch together and share the initial partition, the status messages are single-hop, alerts are still addressed to every UAV whether it is alive or not, and forwarding reads the holdings of the receiver as the summary that would be exchanged on contact. The lease detector is only one of several possible designs, and we have not compared it with the others. Our auction baseline is not a full CBBA, and our GeoSaW-style forwarder uses no planned waypoints. Finally, the 15% reserve rule and the noise levels are assumptions of our scenarios.

TABLE IV
Results after a UAV loss.

Scenario	Cover. %	Time (s)	Failover (s)
5 km, no loss	100.0	353	n/a
5 km, loss	98.4	554	231
3 km, loss	99.0	326	143
1.5 km, loss	99.2	171	102

VIII. Ongoing and Future Work

- Trained single-modality baselines. RGB-only and thermal-only models could be trained next to the fused one, which would replace the masked inference with a proper modality ablation.
- Failure detection. Status messages could carry third-party state, and rendezvous points and estimates of the contact intervals could be used, so that detection does not need to wait for an overdue return.
- Realistic imaging. The static frame of each sector could be replaced by imagery that depends on the

position of the UAV and on a camera model, so that localization becomes geolocation.

- Environment and energy. Richer models of wind, payload and return-trip margin are needed, and a small non-monotonicity of the scaling between 4 and 6 UAVs remains unexplained.

IX. Conclusion

In this paper, we presented a simulated search-and-rescue swarm with early-fusion RGB and thermal perception, a delay-tolerant mesh with store-carry-forward relaying of alerts, and a decentralized controller without a central planner. Buffering was the largest effect that we measured, since without it almost no alert reaches the operator in a spread-out swarm. Failover recovers coverage after a UAV is lost, and forwarding makes alerts faster and safer. The swarm has no central planner, and the price of this is a failover latency of minutes. We also reported the unfavourable results and the limits of a simulation-only study, and the next steps toward real flights are listed in Section VIII.

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